

Errors in Kittel and Kroemer, Thermal Physics, 2nd Edition.

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p. 52. Problem 2. The symbol, “M”, conventionally denotes the total magnetic moment divided by sample volume; here it is total magnetic moment. Also, “logarithm” is misspelled.

p. 68. In Eq. (35), the free energy to be minimized should be defined as:

$$F_0(\tau_R, U, V, N) \equiv U - \tau_R \sigma(U, V, N),$$

where τ_R is the temperature of the thermal reservoir. This free energy is equivalent to the “Landau function” introduced in the Example on p. 69. As defined above, the free energy [Eq. (35)] is a minimum wrt energy transfer between system and reservoir. Note that in Eq. (36), KK take pains to explicitly associate U and σ with the system. By not associating τ with the system, τ is implicitly associated with the reservoir. The “infinitesimal transfer” of some unspecified quantity, [see text just before Eq. (36)], can now be specified as heat flow at fixed reservoir temperature. Of course, the temperature of the system changes when heat flows, but the reservoir temperature does not, consistent with minimizing F_0 wrt U at fixed τ_R . In the text below Eq. (36), the temperature of the system should be defined as $\tau_S^{-1} \equiv \partial\sigma_S/\partial U_S|_{V,N}$. The Helmholtz free energy, $F(\tau, V, N)$, defined by KK in Eq. (35), is F_0 when the system is in equilibrium with the reservoir and their temperatures are the same.

p. 69. In Eqs. (39) and (40), and in the text just below Eq. (40), τ should be τ_R .

p. 81. Problem 2. Should say, “find an exact expression for the equilibrium magnetization, $\langle M \rangle$, and the susceptibility, $\chi \equiv \partial\langle M \rangle/\partial B_e|_{\tau,V,N}$ as functions of temperature and magnetic field... The result for the equilibrium magnetization is $\langle M \rangle = \dots$

p. 132. Near top of page, text should read, “Further, select the ratio of dU to dN in such a way that the total temperature change, $d\tau$, will be zero.” Note: $d\sigma$ follows from the choice of dU and dN , it cannot be chosen independently, as suggested by the original wording.

p. 152. 2nd paragraph should begin, “An **orbital** is a state of the Schrodinger equation for the center-of-mass coordinates of only one particle.” The underlined words are missing from the book. It is clear later that orbitals are particle-in-a-box states, in the simplest treatment of particles.

p. 175. Under the italicized heading, *What is the temperature after expansion?*, the third sentence should read, “The temperature, i.e., kinetic energy, of a real gas may change in the process...”

p. 220. Fig. 7.18. Temperature axis is mK (millikelvin), not K.

p. 264. In going from Eq. (13) to Eq. (14), KK should state that μ_j in Eq. (14) is a function of the partial pressure, p_j , exerted by chemical species j .

p. 262. The argument that G is a minimum for fixed p and τ is weak. Better to start from $\Delta\sigma_{\text{tot}} = \Delta\sigma_R + \Delta\sigma = -\Delta Q/\tau_R + \Delta\sigma$, where ΔQ is the heat removed from the thermal reservoir in order to change the state of the system, and then use conservation of energy to replace ΔQ by $\Delta U + p_R\Delta V$.

p. 330. Problem 11.2 should refer to Fig. 11.7, not Fig. 11.8, for the phase diagram of mixtures of helium-3 and helium-4.

p. 346. Fig. 12.7. The figure indicates that during cooling by adiabatic demagnetization the external magnetic field is turned off at a particular time, and that the entropy and temperature of spin and lattice degrees of freedom continue to decrease at later times. In fact, the magnetic field is turned off slowly, so that the spin degrees of freedom stay in thermal equilibrium. Temperature and entropy change only while the field is changing.

p. 346. Caption to Fig. 12.7 has an extra “s” at the end of the first line.